

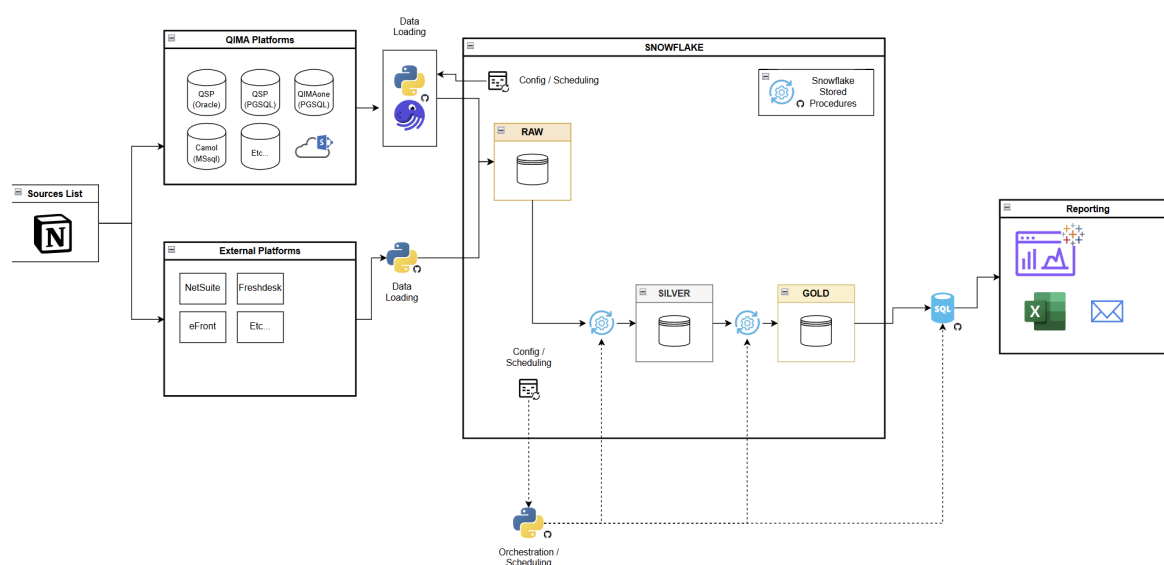


Data lineage — Architecture

Overview

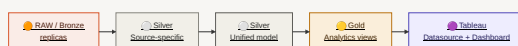
This document describes how data flows from source systems to dashboards in our BI stack.

We use a **medallion architecture** (Bronze (Raw) → Silver → Gold), hosted in Snowflake, with Tableau as the reporting layer.



The **Inspection Analytics** pipeline is used throughout as a concrete example to illustrate each layer.

Architecture summary



Layer	Purpose	What lives here	What doesn't
Raw/Bronze	Raw replicas of source systems, cloned as-is	Data ingestion, availability in Snowflake	No transformation, no cleaning

Layer	Purpose	What lives here	What doesn't
Silver — source-specific	First transformation layer — where the bulk of logic lives	Cleaning, standardization, base metric computation (AQL, SLA flags), coalesce/fallback chains, product hierarchy resolution. Physical tables refreshed via stored procedures.	No cross-source merging
Silver — unified <i>(if applicable)</i>	Merge multiple sources into one canonical schema. Only when multiple systems store the same data type.	Mapping/normalization across sources, row-level security (access management). Each procedure owns its SOURCE slice.	No heavy cleaning — that's done in source-specific silver
Gold	Modelling / shaping based on business needs	Fact-dimension bridging, analytics structuring, fan-outs for multi-granularity. Implemented as views (<code>VW_TBL_*</code>).	No cleaning or minimal only — no business logic
Tableau	Reporting layer	Datasource extracts, dashboards, visual aggregates, filters, formatting	No custom calculation logic, no data cleaning

Snowflake — domain-based structure

The Snowflake environment is organized by **domain** for Silver and Gold layers (cf  [Database](#)) .

Each domain DB contains `*_SILVER` for cleaned/transformed tables and `*_GOLD` for analytics/AI-ready views.

`RAW` is separate — a unified BRONZE database holds raw replicas across all sources, independent of any domain. All sources currently ingested in this location are available via [→| Connection](#)

▼ List of Domain Databases

Database

Aα Name	≡ Note
 FINANCE	Revenue, AR, AP, P&L etc....
 MARKETING	Leads / Email Campaings (Zoho, Marketo)
 CUSTOMER	Customers / Supplier / Factories / Contacts etc.....
 SUPPORT	Freshdesk / Customer Satisfaction Data (NPS)
 HR	Employee data
 PRODUCT	Monitor usage / activity of our different applications and products

Aa Name	≡ Note
 <u>IT</u>	GitHub / Confluence / Logs / System Metrics
 <u>CERTIFICATION</u>	Data related to Certification Services
 <u>INSPECTION</u>	Data related to Inspection Services
 <u>AUDIT</u>	Data related to Audit Services
 <u>LAB_TESTING</u>	Data related to Lab Testing Services
 <u>PROF_SERVICES</u>	Data related to Professional Services
 <u>ANALYTICS</u>	Consolidation of data across business domains (I.e: Inspection + Audit + Lab Testing)
 <u>COMMON</u>	Stores generic dimensions / catalogues / mappings / translations (locations / services / local BU value to Group value etc...)
 <u>ADMIN</u>	Side database used to administrate the data platform and medaillon architecture • procedures • metrics (quality, performance, security) • access mapping (Snowflake User account ↔ HR ID etc... that we currently have under our ACCESS_MNGMT schema)

RAW / Bronze — Source replicas

Mirror source system tables as-is. No transformation, no cleaning. Each Schema corresponds to a schema from the source

Source system	Snowflake location	→ <u>Connection</u>	Notes
QSP / Oracle	<code>RAW.QSP__AIPROD__AIDATAUSER.*</code>	→ <u>QSP__AIPROD - Oracle</u> - <u>aiprod.cqvri9lx1hhi.ap-southeast-1.rds.amazonaws.com</u>	AIPROD (sid) database of QSP, schema AIDATAUSER
QSP / Oracle	<code>RAW.QSP__AIPROD__LTMASTER.*</code>	→ <u>QSP__AIPROD - Oracle</u> - <u>aiprod.cqvri9lx1hhi.ap-southeast-1.rds.amazonaws.com</u>	AIPROD (sid) database of QSP, schema LTMASTER
QIMAone	<code>RAW.QIMAONE__QIMAONE__QIMAONE.*</code>	→ <u>QIMAone - PostgreSQL</u> - <u>qimaone-integration-rwqucro.c5pd69c4wfvw.eu-central-1.rds.amazonaws.com</u>	QIMAone source, QIMAone database, QIMAone schema in that database

Silver — Domain-specific consolidation (CUSTOMER / HR / INSPECTION etc....)

Clean, standardize, and compute base metrics for a **single source system**. This is where the bulk of the transformation logic lives.

Materialized as physical tables, refreshed via snowflake stored procedures. Procedures reside in the Silver schema and are version-controlled in Git.

When multiple source systems store the same type of information (e.g. QSP and QIMAone both produce inspections), this step handles each system independently — one procedure per source. For domains with a single source, this is the only silver step before gold.

Inspection Analytics example — QSP source:

Table	<code>INSPECTION.CP__SILVER.QSP_REPORT</code>
Procedure	<code>REFRESH_QSP_REPORT</code> (GitHub)
Feeds from	~15 raw tables from <code>RAW.QSP__AIPROD__AIDATAUSER.*</code>
Grain	One row per report

Silver — unified model

Merge source-specific silver tables into a **single canonical schema**. This step only applies when multiple sources feed the same domain.

Each procedure owns its `SOURCE` slice — deletes and re-inserts. Row-level security (access management) is also computed at this stage.

Inspection Analytics example:

Table	<code>INSPECTION.CP__SILVER.REPORT</code>
Procedures	<code>REFRESH_REPORT__QSP()</code> (QSP slice) + <code>REFRESH_REPORT__QIMAONE()</code> (QIMAone slice)
Grain	One row per report

Cross-domain dependencies

The unified silver layer typically joins with other domains' tables for mapping, access and reference data:

Domain	 Database	What it provides
<code>COMMON.MAPPINGS</code>	 <u>COMMON</u>	Reference data — country-cleaning, locations, AQL levels
<code>COMMON.MASTER</code>	 <u>COMMON</u>	Master Data / Standard ontologies
<code>ADMIN.ACCESS_MNGMT</code>	 <u>ADMIN</u>	Row-level security rules / policies

Gold — analytics & AI-ready views

Shape silver tables for specific dashboard / analytics needs. Joins facts with dimensions. Minimal transformation — mostly bridging and aliasing.

Gold views are the interface between Snowflake and the reporting layer. They are designed to be consumed directly by Tableau datasources.

Naming convention: Views prefixed with `VW_TBL_` are designed for direct Tableau consumption.

Inspection Analytics example:

View	<code>INSPECTION.CP_GOLD.VW_TBL_CLIENT_ANALYTICS</code>
Location	<code>INSPECTION.CP_GOLD</code>
Main input	<code>INSPECTION.CP_SILVER.REPORT / INSPECTION.CP_SILVER.REPORT_CHECKPOINT / INSPECTION.CP_SILVER.REPORT_DEFECT etc....</code>

This view bridges the unified inspection fact table with dimensions (company, entity, inspector, benchmarks, custom fields) and uses a cross-join fan-out to produce multiple row types per inspection. This lets a single Tableau extract serve multiple granularities without needing separate datasources.

Reporting — Tableau

Data Sources

Selects the gold view and materializes it as a Tableau extract. Minimal calculation layer — mostly for shaping data to fit visualization needs.

One gold view typically maps to one Tableau datasource - though some Tableau datasources can also be composed / linked to several views to also leverage Tableau dynamic relations features.

Inspection Analytics example: [datasource/82525](#)



Inspection Analytics

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Owner

Clement Crevecoeur

Data is

Extract

▼ Jun 20, 2026, 5:14 AM

[GitHub Link](#)

New ▼

Edit Data Source

Connections 1

Extract Refreshes 1

Connected Workbooks 17

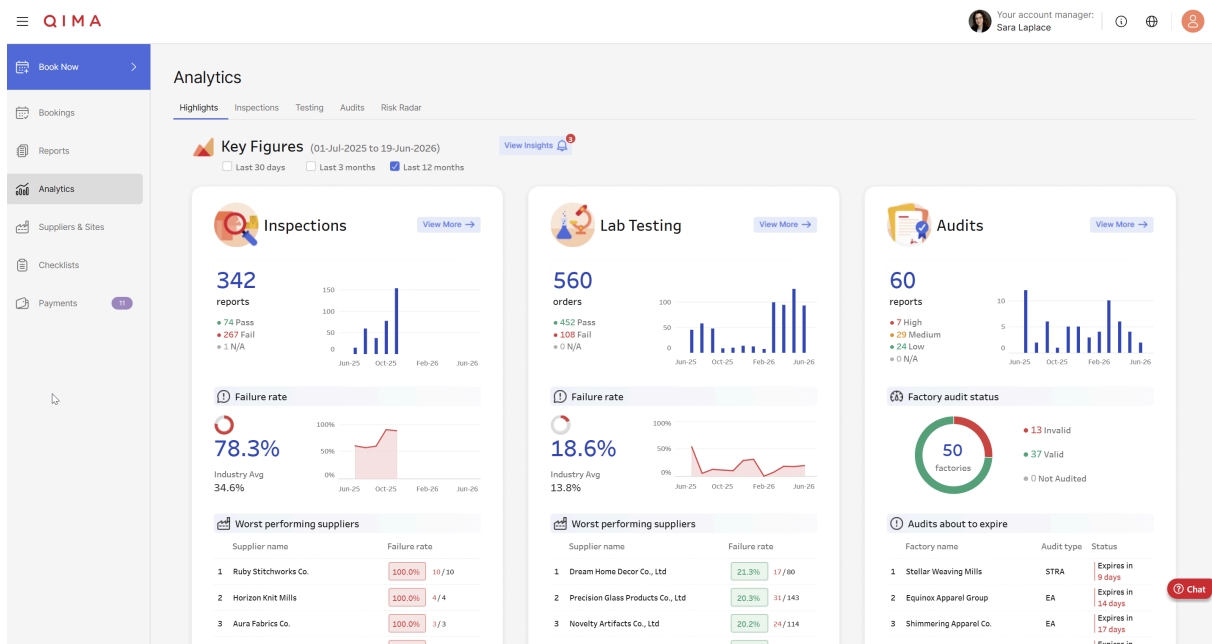
Select All

	Name	Actions	Views (all-time)	Sheets	Size
☐	☆ ACA - Highlights	⋮	187,917	4	1.5 MB
☐	☆ ACA - Inspection Analytics	⋮	110,769	18	2.3 MB
☐	☆ QIMAone Analytics - Highlights	⋮	24,271	1	821.7 KB
☐	☆ QIMAone Analytics - Inspections	⋮	18,730	18	2.7 MB

Dashboards

Pure presentation and aggregation. One datasource can feed multiple dashboard tabs / workbooks.

Inspection Analytics example: Inspection Analytics — HIGHLIGHTS



Summary — where does each type of logic live?

Logic type	Layer
Raw data replication	Bronze
Data cleaning, standardization, base metrics	Silver (source-specific)

Logic type	Layer
Source unification, row-level security	Silver (unified)
Dimension bridging, analytics shaping	Gold
Visual aggregation, presentation	Tableau

Example — Snowflake Lineage

Snowflake also automatically generate lineage and give an overview on downstream & upstream dependencies for a given object.

